

INVESTMENTS

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CHAPTER 7

Optimal Risky Portfolios

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · Diversification and Portfolio Risk

The investment decision is a top-down process: (1) capital allocation between the risky portfolio and the risk-free asset (Chapter 6), (2) asset allocation within the risky portfolio across broad asset classes, and (3) security selection of individual assets within each class. Asset allocation and security selection are technically the same problem — both seek the combination of risky assets offering the best risk–return trade-off, i.e., the highest Sharpe ratio — but are usually separated in practice for convenience.

Market Risk vs. Firm-Specific Risk

As more securities are added to a portfolio, risk generally falls, but not to zero. **Firm-specific (unique/diversifiable) risk** — shocks that affect only one company (e.g., a lawsuit, a factory fire) — tends to average out and can be diversified away almost completely in a large portfolio. **Market (systematic/non-diversifiable) risk** — factors that affect the whole economy (interest rates, GDP growth) — affects all firms simultaneously and cannot be eliminated by adding more securities.

As portfolio size (n) grows, firm-specific risk shrinks toward zero, but total portfolio risk converges toward the level of market risk alone — diversification has a floor, not a path to zero risk.

LO2 · Portfolios of Two Risky Assets

For a portfolio of a bond fund (D) and an equity fund (E), expected return is a simple weighted average, but portfolio **variance** depends critically on the **covariance (or correlation)** between the two assets' returns (see Formulas page).

The Diversification Effect

Because portfolio standard deviation is less than the weighted average of the component standard deviations whenever correlation is below +1, combining imperfectly correlated assets **always** provides some diversification benefit. A **hedge asset** (negative correlation with the rest of the portfolio) is especially effective at reducing risk, since expected return is unaffected by correlation — only risk is. In the extreme case of perfect negative correlation ($\rho = -1$), portfolio weights exist that drive standard deviation all the way to zero.

The Minimum-Variance Portfolio

For any two risky assets, a unique combination minimizes portfolio variance (see Formulas page). This minimum-variance portfolio always has a standard deviation **lower than either individual asset's** — the clearest illustration of the diversification effect. As the portfolio weight in the higher-risk asset rises from zero, standard deviation typically falls at first (diversification benefit dominates) before rising again once the portfolio becomes too concentrated in the riskier asset.

LO3 · Asset Allocation with Stocks, Bonds, and Bills

Adding a risk-free asset (T-bills) to the two-risky-asset problem lets us find the single best combination of the risky assets — the **optimal risky portfolio, P**. Among all possible risky-asset combinations, P is the one whose combination with T-bills produces the **steepest possible Capital Allocation Line**, i.e., the highest attainable Sharpe ratio (see Formulas page for the weight formula).

Once P is identified, every investor — regardless of risk aversion — combines it with T-bills using exactly the same y^* formula from Chapter 6. This is the **separation property**: the choice of the optimal risky portfolio is a purely technical problem (the same for all investors), fully separate from the personal, preference-driven choice of how much of that risky portfolio to hold (which varies by investor risk aversion).

Separation property: one optimal risky portfolio serves every investor; only the allocation between it and the risk-free asset differs by client.

LO4 · The Markowitz Portfolio Optimization Model

Harry Markowitz (1952 paper, 1990 Nobel Prize) generalized the two-asset problem to **n risky assets** plus a risk-free asset. The procedure has three steps:

- **1. Risk-return analysis** — using n estimates of expected return and $n(n-1)/2$ distinct covariance estimates, plot the minimum-variance frontier: the lowest attainable variance for every level of expected return. Portfolios on this frontier above the global minimum-variance portfolio form the efficient frontier.
- **2. Find the optimal risky portfolio** — search among efficient-frontier portfolios for the one whose CAL (tangent line from the risk-free rate) has the steepest slope — the highest Sharpe ratio.
- **3. Choose the complete portfolio** — combine the optimal risky portfolio with T-bills according to each investor's risk aversion (exactly as in Chapter 6).

The data burden grows quickly: a 50-security portfolio requires 50 expected-return estimates, 50 variance estimates, and $50 \times 49/2 = 1,225$ distinct covariance estimates — a major practical obstacle that motivates the simplified index models of Chapter 8.

LO5 · Risk Pooling, Risk Sharing, and the Risk of Long-Term Investments

A subtle but important distinction underlies the common intuition that "diversification reduces risk":

Risk Pooling

Adding more uncorrelated bets to a portfolio **while increasing the total amount invested** (e.g., an insurer selling more policies) makes the *rate of return* more predictable, but total **dollar risk actually increases** — just as a gambler making more roulette bets faces more total dollar risk even as their per-bet average becomes more predictable. Risk pooling alone does not reduce risk; it merely changes how that (growing) risk is measured.

Risk Sharing

The real risk reduction comes from **risk sharing**: keeping your total invested amount fixed while spreading exposure across more assets and letting other investors share in each position (e.g., selling off half your stake as you add a new asset). Only when the total investment base is held constant does adding uncorrelated assets genuinely lower risk. Capital markets provide exactly this mechanism, letting large pools of investors each hold a small, diversified stake without any single investor needing to increase their total capital at risk.

Implication for Long-Term Investing

Extending an investment's time horizon is analogous to risk pooling, not risk sharing — stringing together more periods of exposure without reducing the total capital at risk in any one of them. The popular claim that "stocks are safer over long horizons" (time diversification) is therefore **not automatically true**; the actual dollar risk to an investor's terminal wealth can grow with the investment horizon, since more periods of uncertain returns compound rather than cancel out.

Chapter 7 — Big-Picture Takeaways

- Firm-specific risk can be diversified away almost entirely; market (systematic) risk cannot, no matter how many securities are added to a portfolio.
- Combining assets with correlation below +1 always provides some diversification benefit; the minimum-variance portfolio has lower risk than either individual asset.
- The optimal risky portfolio is the one combination of risky assets that maximizes the Sharpe ratio (steepest CAL) — the same for every investor (the separation property).
- The Markowitz model generalizes this to n assets via the efficient frontier, at the cost of a rapidly growing number of required covariance estimates.
- True risk reduction requires both risk pooling AND risk sharing; risk pooling alone (more bets, more capital at risk) can actually increase total dollar risk.
- Long-horizon investing is not automatically "safer" — extending the horizon resembles risk pooling more than risk sharing.

These formulas extend the two-asset portfolio math into the core toolkit used for the rest of the course.

F1 | Two-Asset Portfolio — Expected Return

$$E(r_P) = w_D \times E(r_D) + w_E \times E(r_E)$$

Simple weighted average of the two component expected returns. $w_D + w_E = 1$.

F2 | Two-Asset Portfolio — Variance

$$\sigma_P^2 = w_D^2 \sigma_D^2 + w_E^2 \sigma_E^2 + 2 w_D w_E \text{Cov}(r_D, r_E)$$

$$\text{Cov}(r_D, r_E) = \rho_{DE} \times \sigma_D \times \sigma_E$$

Unlike expected return, variance is NOT a simple weighted average — the covariance (or correlation) term is what creates the diversification effect. Lower $\rho \rightarrow$ lower σ_P for given weights.

F3 | Minimum-Variance Portfolio Weights (Two Assets)

$$w_D(\min) = [\sigma_E^2 - \text{Cov}(r_D, r_E)] / [\sigma_D^2 + \sigma_E^2 - 2\text{Cov}(r_D, r_E)]$$

$$w_E(\min) = 1 - w_D(\min)$$

The unique weighting that minimizes σ_P^2 . Found by setting the derivative of F2 with respect to w_D to zero and solving.

F4 | Optimal Risky Portfolio Weights (Two Risky Assets + T-Bills)

$$w_D^* = [E(R_D)\sigma_E^2 - E(R_E)\text{Cov}(R_D, R_E)] / [E(R_D)\sigma_E^2 + E(R_E)\sigma_D^2 - (E(R_D) + E(R_E))\text{Cov}(R_D, R_E)]$$

$$w_E^* = 1 - w_D^*$$

R denotes EXCESS returns (over r_f), not total returns — a common point of confusion. This weighting maximizes the Sharpe ratio of the risky portfolio, making its CAL the steepest possible.

F5 | Generalization to n Assets (Markowitz Model)

$$E(r_P) = \sum w_i E(r_i)$$

$$\sigma_P^2 = \sum \sum w_i w_j \text{Cov}(r_i, r_j) \quad (\text{sum over all } i, j \text{ pairs, including } i = j \text{ where } \text{Cov}(r_i, r_i) = \sigma_i^2)$$

Direct extension of F1–F2 to any number of assets. Requires n expected returns, n variances, and $n(n-1)/2$ distinct covariances as inputs.

EXAMPLES 7.1–7.3 | Bond & Stock Fund — Full Solution

Given (Table 7.1): Bond fund D: $E(r)=8\%$, $\sigma=12\%$. Stock fund E: $E(r)=13\%$, $\sigma=20\%$. $\text{Cov}(D,E)=72$ ($\rho=0.30$). Risk-free rate $r_f = 5\%$.

Step 1 — Minimum-Variance Portfolio:

$$w_D = 0.82, w_E = 0.18 \rightarrow \sigma_{\min} = \sqrt{(.82^2 \times 144) + (.18^2 \times 400) + (2 \times .82 \times .18 \times 72)} = 11.45\%$$

Note this is lower than EITHER fund's own standard deviation (12% and 20%) — the diversification effect in action.

Step 2 — Optimal Risky Portfolio, P:

$$w_D = 0.40, w_E = 0.60$$

$$E(r_P) = .4(8) + .6(13) = 11\%$$

$$\sigma_P = \sqrt{(.4^2 \times 144) + (.6^2 \times 400) + (2 \times .4 \times .6 \times 72)} = 14.2\%$$

$$\text{Sharpe ratio: } S_P = (11 - 5) / 14.2 = 0.42$$

Step 3 — Optimal Complete Portfolio (investor with $A = 4$):

$$y^* = (11 - 5) / (4 \times 14.2^2) = 6 / 806.6 = 0.7439$$

→ 74.39% in portfolio P, 25.61% in T-bills

$$\begin{array}{ll} \text{Fraction in bonds} = .7439 \times .40 = 29.76\% & | \quad \text{Fraction in stocks} \\ = .7439 \times .60 = 44.63\% \end{array}$$

Notice the three-stage logic: first find the best risky mix (P), independent of any investor's preferences; then let each investor's risk aversion determine how much of P (versus T-bills) to hold — this is the separation property from LO3.

PRACTICE PROBLEM | Rundle Peak Capital

Scenario

You are an analyst at Rundle Peak Capital in Calgary, building a two-asset portfolio recommendation for a client. Your firm covers a Canadian bond fund (B) and a Canadian equity fund (S) with the following characteristics:

Fund	$E(r)$	σ	Correlation
Bond Fund (B)	6%	10%	$\rho = 0.20$
Stock Fund (S)	14%	24%	(with Bond Fund)

Additional info: The risk-free rate is 4%. A client has a coefficient of risk aversion $A = 3.5$.

Part A — Covariance and Portfolio Risk

- Calculate the covariance between the bond and stock funds from the correlation and standard deviations given.
- Calculate the expected return and standard deviation of a portfolio with 60% in bonds and 40% in stocks.
- Explain, without calculating, why this portfolio's standard deviation is less than a simple weighted average of 10% and 24%.

Part B — Minimum-Variance Portfolio

- Calculate the minimum-variance portfolio weights for bonds and stocks.
- Calculate the standard deviation of this minimum-variance portfolio.
- Confirm that this standard deviation is lower than either fund's individual standard deviation.

Part C — Optimal Risky Portfolio

- Calculate the optimal risky portfolio weights, w_{B^*} and w_{S^*} , using the excess-return formula.
- Calculate the expected return and standard deviation of this optimal risky portfolio, P.
- Calculate the Sharpe ratio of portfolio P.

Part D — Optimal Complete Portfolio

- Calculate y^* , the optimal fraction of the client's wealth to place in portfolio P, given $A = 3.5$.
- Calculate the resulting fraction of total wealth in the bond fund and in the stock fund.

(iii) A second, more risk-averse client has $A = 7$. Without recalculating portfolio P, explain what changes (and what stays the same) in this client's solution, referencing the separation property.

Part E — Risk Pooling vs. Risk Sharing

Rundle Peak is also advising an insurance subsidiary that wants to reduce risk by underwriting more, uncorrelated policies without bringing in additional outside investor capital.

(i) Explain why this strategy, by itself, will not reduce the subsidiary's total dollar risk.

(ii) What additional step (in terms of LO5) would allow the subsidiary to actually reduce risk while writing more policies?

Hints & Guidance

Part A tests LO2 and Formulas F1–F2. Part B tests LO2 and Formula F3. Part C tests LO3 and Formula F4 (mirror Examples 7.1–7.2 exactly). Part D tests LO3 and the Chapter 6 y^ formula — remember only y^* changes across clients, not portfolio P itself. Part E tests LO5 — the key distinction is risk pooling (more bets, same or more capital at risk) vs. risk sharing (spreading a FIXED capital base across more investors).*